National University of Computer and Emerging Sciences

Lahore Campus

Data Structures Lab (CL2001)

BCS-3H

Midterm Exam

Date: October 15, 2026

Total Time (Hrs): 2

Total Marks: 25

Total Questions: 2

Instructor

Mr. Jawad Khan

Submission Path

Windows + R: \\Exam\ Exam\Fall 2026 Midterm Exam\ DS Lab\ [your section]\ Submissions

Instructions:

- Duration of exam is 2 hours. An additional 15 minutes have been allocated for proper submission of your files. Therefore, stop attempting the exam once the 2 hour duration is over and use the remaining time to carefully submit your file.
- Make a .cpp file for each question and name it using your roll number e.g. 25L-1234-q1.cpp
- Place all the exam (.cpp) files in a single folder and compress the folder into a ZIP file named using your roll number e.g. 25L-1234.zip and upload the ZIP file to the designated submission folder.
- After submitting, verify your zip file is visible in the submission folder of your section, and check that a file size is shown next to it (to ensure it is not 0 KB file which indicates the file is empty or corrupted).
- To ensure everything is properly submitted, you may download the uploaded ZIP file and check its contents. It is your responsibility to save and submit all required files in the correct format and ensure that the submitted files are complete and not corrupted.
- Submit your files inside your section's submissions folder

Attempt all the questions.

CLO #1,2

Question 1 **[15 Marks]**

Problem: Recursive Flood Fill and State Recovery

Implement a **Stack using a Singly Linked List** where each node holds (row, col, old_color). Write a **recursive function** flood_fill_recursive(...) that performs a bucket fill on an $M \times N$ matrix starting from a given target pixel.

- **Requirement:** Every modified pixel's old state must be pushed onto your Linked List Stack during recursion. Write a secondary undo() function that pops from your stack to restore the grid to its initial state.

Example:

- **Initial 3 x 3 Grid:**

[1, 1, 2]

[1, 1, 0]

[2, 0, 1]

- **Fill Parameters:** Start at (0, 0), New Color = 5
- **Grid After Flood Fill:**

[5, 5, 2]

[5, 5, 0]

[2, 0, 1]

- **Action:** Call undo() function (Pops all saved states from Stack)
- **Expected Output:** Grid reverts completely back to original matrix values.

CLO #1,2

Question 2 [10 marks]

Problem: Next Greater Element using Stack

You are given an array of integers representing stock prices over consecutive days.

- **Requirement:** For every element, find the first element to its right that is greater than it. Use a **Stack implemented with a Singly Linked List (Implemented in Q1)** to store indices or values. If no greater element exists,

  output -1.
- **Time Constraint:** Algorithm must process the array in **O(N) total time complexity**. A nested loop solution O(N^2) will lose marks.

Example:

- **Input Array:** [4, 5, 2, 25, 10]
- **Processing:**
  - For 4 > Next greater is 5
  - For 5 > Next greater is 25
  - For 2 > Next greater is 25
  - For 25 > No greater element > -1
  - For 10 > No greater element > -1
- **Expected Output:** [5, 25, 25, -1, -1]

---------- Good luck! ----------